

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
.....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
.....
COMMISSION NATIONALE D'ORGANISATION DE
L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)
.....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
.....
MINISTRY OF HIGHER EDUCATION
.....
NATIONAL COMMISSION FOR THE ORGANISATION
OF HIGHER NATIONAL DIPLOMA EXAM (HND)
EXAM
.....

National Exam of Higher National Diploma 2025 Session

Spécialty/option: Software Engineering
Paper : System Analysis and Design
Duration : 4 hours

Credits : 7

GENERAL INSTRUCTIONS.

Answer All Questions

*You are reminded of the necessity of good English and orderly presentation of your material.
Where calculations are required, clearly show your working and be chronological in your answer.*

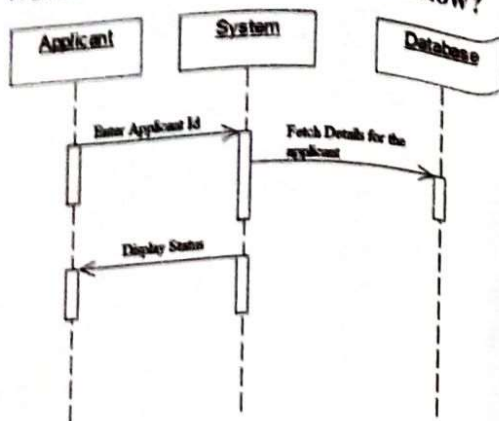
SECTION A: INFORMATION SYSTEMS AND DATABASE. (50 MARKS)

Part I: Object Modeling (25 Marks)

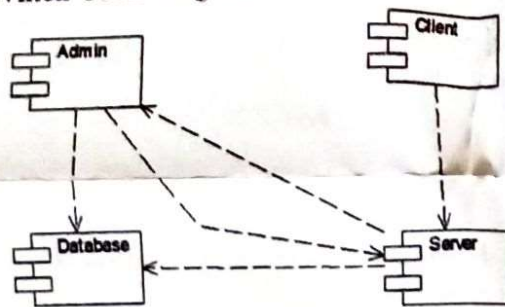
MCQs: One (01) Mark for Each Correct Answer. Write Your Answers in Your Booklet

1. For the success of project development Requirements analysis is the critical stage
 - a) Yes
 - b) No
 - c) Based on the size of project
 - d) None of the above
2. _____ does not correspond to a good Software Requirement Specification (SRS)
 - a) Verifiable
 - b) Complete
 - c) Traceable
 - d) Ambiguous
3. Which UML diagram supports event-based modeling?
 - a) Deployment
 - b) Collaboration
 - c) State chart
 - d) All of the above
4. Which of the following UML diagrams represent interaction modeling?
 - a) Use Case, Sequence
 - b) Class, Object
 - c) Activity, State Chart

- d) All of the above
5. Which UML diagram is shown below?



- a) Use Case
 b) Collaboration
 c) Sequence
 d) Object
6. Which UML diagram is shown below?



- a) Component
 b) Deployment
 c) Use Case
 d) Class
7. Which of the following is a structural model that demonstrates the other systems in the environment of the system being developed?
- a) System context model
 b) Interaction model
 c) Environmental model
 d) both a and b
8. Which model shows the flow of object interactions?
- a) Sequence model
 b) Subsystem model
 c) Dynamic model
 d) Structural model

9. Which diagram indicates the behavior of the system as a consequence of external events?
- a) Data flow
 - b) State transition
 - c) Control specification
 - d) Work flow
10. _____ defines the properties of a data object and take on one of the different characteristics
- a) Data object
 - b) Attributes
 - c) Relationships
 - d) All of the above
11. The _____ enables the software engineers to develop models of the information domain and functional domain at the same time
- a) Data flow diagram
 - b) State transition diagram
 - c) Control specification diagram
 - d) Activity diagram
12. Which of the following is not a construct?
- a) Sequence
 - b) Condition
 - c) Repetition
 - d) System
13. The _____ is represented as a rectangular box in flow chart
- a) Repetition
 - b) Sequence
 - c) Condition
 - d) None of the above
14. Which granularity level of testing checks the behavior of module cooperation?
- a) Unit Testing
 - b) Integration Testing
 - c) Acceptance Testing
 - d) Regression Testing
15. Which of the following is a design pattern?
- a) Behavioral
 - b) Structural
 - c) Abstract Factory
 - d) All of the above
16. Classes and interfaces are a part of
- a) Structural things
 - b) Behavioral things
 - c) Grouping things
 - d) Annotational things



17. What is a physical element that exists at runtime in UML?
- a) A node
 - b) An interface
 - c) An activity
 - d) None of the above
18. Which things are dynamic parts of UML models?
- a) Structural things
 - b) Behavioral things
 - c) Grouping things
 - d) Annotational things
19. If you are working on real-time process control applications or systems that involve concurrent processing, you would use a
- a) Activity diagram
 - b) Sequence diagram
 - c) Statechart diagram
 - d) Object diagram
20. Which things in UML are the explanatory parts of UML models?
- a) Structural things
 - b) Behavioral things
 - c) Grouping things
 - d) Annotational things
21. What refers to the value associated with a specific attribute of an object and to any actions or side?
- a) Object
 - b) State
 - c) Interface
 - d) None of the mentioned
22. Select the statement true for activity diagrams.
- a) They can be used to discover parallel activities
 - b) They are used to depict workflow for a particular business activity
 - c) Activity diagram do not tell who does what and are difficult to trace back to object models
 - d) All of the above
23. Which of the following are the valid relationships in Use Case Diagrams
- a) Generalization
 - b) Include
 - c) Extend
 - d) All of the above
24. UML interfaces are used to:
- a) specify required services for types of objects
 - b) program in Java, but not in C++ or Smalltalk
 - c) define executable logic to reuse across classes
 - d) define an API for all classes

25. How many diagrams are there in Unified Modelling Language?

- a) six
- b) seven
- c) eight
- d) nine

Part II: Data Base: (25 Marks)

1) Based on the given table, solve the following queries.

Employee table

empId	empName	Age	address
1001	Bachi	26	Delhi
1002	Hamad	30	Gurgaon
1003	Yousouf	27	Mumbai
1004	Rafa	32	Nagpur

- a) Write the SELECT command to display the details of the employee with empId as 1004. (2 mark)
 - b) Write the SELECT command to display all the records of table Employees. (2 marks)
 - c) Write the SELECT command to display all the records of the employee whose name starts with the character 'R'. (3 marks)
- 2) As a software engineer, you have been awarded a contract to produce a database software for a certain University in town. In the system specification, you are told that the database will have to deal with:
- STUDENTS who have Student ID, First Name, Last Name, Date of Birth
 - DEPARTMENTS which have department Code, Department Title, Faculty
 - DEGREE which constitute Degree Code, Degree Title, Department Code
 - COURSE which include Course No, Course Code, Course Title, No of hours
 - REGISTER which have Registration No, Degree Code, Student No, Year of entry
 - REQUIRES which have Requirement No, Course No, Degree Code

Tasks:

- a) List entities in the case study above (2 marks)
 - b) List the attributes for DEGREE (1 mark)
 - c) Design an Entity Relationship (ER) Diagram for the above system indicating multiplicity (1 mark)
 - d) Present the conceptual model of the system (2 marks)
 - e) Present the Logical model of the system (2 marks)
 - f) Present the physical model of the system (2 marks)
- 3) What is Data Integrity in database? (2 marks)
- 4) What are Entities and Relationships? (3 marks)
- 5) What is an Alias in SQL? (2 marks)
- 6) What are the various forms of Normalization? (1 mark)

SECTION B: WEB AND MOBILE PROGRAMMING: (50 MARKS)

Part I: Web Programming (25 Marks)

- 1) What are the limitations of CSS? (3 marks)
- 2) Write the basic structure of the HTML template? (3 marks)
- 3) Do all HTML tags have an end tag? (2 marks)
- 4) What are the uses of HTML5 (3 marks)
- 5) Write an HTML5 code to form a table to show the below values in a tabular form with heading as Roll No., Student name, Subject Name, and values as follows:
 - a) Ram, Physics (2 marks)
 - b) Shyam, Math (2 marks)
 - c) Murli, Chemistry (2 marks)
- 6) Differentiate between HTML and XHTML. (4 marks)
- 7) Name two new tags included in the HTML 5 (2 marks)
- 8) What are the two types of Web Storage in HTML5? (2 marks)

Part I: Mobile Programming (25 Marks)

- 1) What is a mobile operating system? (3 marks)
- 2) Explain four (04) Android Architectures. (2 marks)
- 3) When is the best time to kill a foreground activity? (2 marks)
- 4) Explain what happens at each of the following layers of mobile operating systems:
 - a) Hardware layer. (2 marks)
 - b) Mobile OS layer. (2 marks)
 - c) Applications layer. (2 marks)
- 5) With the help of an example in each case, explain the uses of the following mobile operating systems:
 - a) iOS *Apple* (3 marks)
 - b) Android *google* (3 marks)
 - c) Windows 10 mobile *microsoft* (3 marks)
 - d) Ubuntu Touch (3 marks)